In [3]:
```
pip install mlxtend
```

```
Collecting mlxtend
  Downloading mlxtend-0.23.4-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: scipy>=1.2.1 in c:\users\student\anaconda3\lib\site-packages (from mlxtend) (1.15.3)
Requirement already satisfied: numpy>=1.16.2 in c:\users\student\anaconda3\lib\site-packages (from mlxtend) (2.1.3)
Requirement already satisfied: pandas>=0.24.2 in c:\users\student\anaconda3\lib\site-packages (from mlxtend) (2.2.3)
Requirement already satisfied: scikit-learn>=1.3.1 in c:\users\student\anaconda3\lib\site-packages (from mlxtend) (1.6.1)
Requirement already satisfied: matplotlib>=3.0.0 in c:\users\student\anaconda3\lib\site-packages (from mlxtend) (3.10.0)
Requirement already satisfied: joblib>=0.13.2 in c:\users\student\anaconda3\lib\site-packages (from mlxtend) (1.4.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\student\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend) (1.3.1)
Requirement already satisfied: cycler>=0.10 in c:\users\student\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\student\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend) (4.55.3)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\student\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend) (1.4.8)
Requirement already satisfied: packaging>=20.0 in c:\users\student\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend) (24.2)
Requirement already satisfied: pillow>=8 in c:\users\student\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend) (11.1.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\student\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\student\anaconda3\lib\site-packages (from matplotlib>=3.0.0->mlxtend) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\student\anaconda3\lib\site-packages (from pandas>=0.24.2->mlxtend) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\student\anaconda3\lib\site-packages (from pandas>=0.24.2->mlxtend) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\student\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib>=3.0.0->mlxtend) (1.17.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\student\anaconda3\lib\site-packages (from scikit-learn>=1.3.1->mlxtend) (3.5.0)
Downloading mlxtend-0.23.4-py3-none-any.whl (1.4 MB)
   ---------------------------------------- 0.0/1.4 MB ? eta -:--:--
   ---------------------------------------- 1.4/1.4 MB 18.7 MB/s eta 0:00:00
Installing collected packages: mlxtend
Successfully installed mlxtend-0.23.4
Note: you may need to restart the kernel to use updated packages.
```

In [4]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
import matplotlib.pyplot as plt


from sklearn import metrics
from sklearn.metrics import confusion_matrix
from mlxtend.plotting import plot_confusion_matrix
```

In [7]:
```
dataset_path=r"C:\Users\Student\Downloads\cancer - cancer.csv"
df = pd.read_csv(dataset_path)
```

In [8]:
```
df
```

Out[8]:

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	...	radius_worst	texture_worst	perimeter_worst	area_worst	smoothness_worst	compactness_worst	concavity_worst	concave points_worst	symmetry_worst
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.30010	0.14710	...	25.380	17.33	184.60	2019.0	0.16220	0.66560	0.7119	0.2654	0.4601
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.08690	0.07017	...	24.990	23.41	158.80	1956.0	0.12380	0.18660	0.2416	0.1860	0.2750
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.19740	0.12790	...	23.570	25.53	152.50	1709.0	0.14440	0.42450	0.4504	0.2430	0.3613
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.24140	0.10520	...	14.910	26.50	98.87	567.7	0.20980	0.86630	0.6869	0.2575	0.6638
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.19800	0.10430	...	22.540	16.67	152.20	1575.0	0.13740	0.20500	0.4000	0.1625	0.2364
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
564	926424	M	21.56	22.39	142.00	1479.0	0.11100	0.11590	0.24390	0.13890	...	25.450	26.40	166.10	2027.0	0.14100	0.21130	0.4107	0.2216	0.2060
565	926682	M	20.13	28.25	131.20	1261.0	0.09780	0.10340	0.14400	0.09791	...	23.690	38.25	155.00	1731.0	0.11660	0.19220	0.3215	0.1628	0.2572
566	926954	M	16.60	28.08	108.30	858.1	0.08455	0.10230	0.09251	0.05302	...	18.980	34.12	126.70	1124.0	0.11390	0.30940	0.3403	0.1418	0.2218
567	927241	M	20.60	29.33	140.10	1265.0	0.11780	0.27700	0.35140	0.15200	...	25.740	39.42	184.60	1821.0	0.16500	0.86810	0.9387	0.2650	0.4087
568	92751	B	7.76	24.54	47.92	181.0	0.05263	0.04362	0.00000	0.00000	...	9.456	30.37	59.16	268.6	0.08996	0.06444	0.0000	0.0000	0.2871

569 rows × 32 columns

In [9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 32 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   id                       569 non-null    int64  
 1   diagnosis                569 non-null    object 
 2   radius_mean              569 non-null    float64
 3   texture_mean             569 non-null    float64
 4   perimeter_mean           569 non-null    float64
 5   area_mean                569 non-null    float64
 6   smoothness_mean          569 non-null    float64
 7   compactness_mean         569 non-null    float64
 8   concavity_mean           569 non-null    float64
 9   concave points_mean      569 non-null    float64
 10  symmetry_mean            569 non-null    float64
 11  fractal_dimension_mean   569 non-null    float64
 12  radius_se                569 non-null    float64
 13  texture_se               569 non-null    float64
 14  perimeter_se             569 non-null    float64
 15  area_se                  569 non-null    float64
 16  smoothness_se            569 non-null    float64
 17  compactness_se           569 non-null    float64
 18  concavity_se             569 non-null    float64
 19  concave points_se        569 non-null    float64
 20  symmetry_se              569 non-null    float64
 21  fractal_dimension_se     569 non-null    float64
 22  radius_worst             569 non-null    float64
 23  texture_worst            569 non-null    float64
 24  perimeter_worst          569 non-null    float64
 25  area_worst               569 non-null    float64
 26  smoothness_worst         569 non-null    float64
 27  compactness_worst        569 non-null    float64
 28  concavity_worst          569 non-null    float64
 29  concave points_worst     569 non-null    float64
 30  symmetry_worst           569 non-null    float64
 31  fractal_dimension_worst  569 non-null    float64
dtypes: float64(30), int64(1), object(1)
memory usage: 142.4+ KB
```

In [10]:
```
df.head()
```

Out[10]:

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	...	radius_worst	texture_worst	perimeter_worst	area_worst	smoothness_worst	compactness_worst	concavity_worst	concave points_worst	symmetry_worst	f
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	...	25.38	17.33	184.60	2019.0	0.1622	0.6656	0.7119	0.2654	0.4601	
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	...	24.99	23.41	158.80	1956.0	0.1238	0.1866	0.2416	0.1860	0.2750	
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	...	23.57	25.53	152.50	1709.0	0.1444	0.4245	0.4504	0.2430	0.3613	
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	...	14.91	26.50	98.87	567.7	0.2098	0.8663	0.6869	0.2575	0.6638	
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	...	22.54	16.67	152.20	1575.0	0.1374	0.2050	0.4000	0.1625	0.2364	

5 rows × 32 columns

In [12]:
```
df['diagnosis'].unique()
```

Out[12]:
```
array(['M', 'B'], dtype=object)
```

In [14]:
```
from sklearn import preprocessing
label_encoder = preprocessing.LabelEncoder()

df["diagnosis"] = label_encoder.fit_transform(df['diagnosis'])
df['diagnosis']
```

Out[14]:
```
0      1
1      1
2      1
3      1
4      1
      ..
564    1
565    1
566    1
567    1
568    0
Name: diagnosis, Length: 569, dtype: int64
```

In [15]:
```
df.columns
```

Out[15]:
```
Index(['id', 'diagnosis', 'radius_mean', 'texture_mean', 'perimeter_mean',
       'area_mean', 'smoothness_mean', 'compactness_mean', 'concavity_mean',
       'concave points_mean', 'symmetry_mean', 'fractal_dimension_mean',
       'radius_se', 'texture_se', 'perimeter_se', 'area_se', 'smoothness_se',
       'compactness_se', 'concavity_se', 'concave points_se', 'symmetry_se',
       'fractal_dimension_se', 'radius_worst', 'texture_worst',
       'perimeter_worst', 'area_worst', 'smoothness_worst',
       'compactness_worst', 'concavity_worst', 'concave points_worst',
       'symmetry_worst', 'fractal_dimension_worst'],
      dtype='object')
```

In [16]:
```
df.describe()
```

Out[16]:

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	...	radius_worst	texture_worst	perimeter_worst	area_worst	smoothness_worst	compactness_worst	concavity_worst	concave points_worst	sym
count	5.690000e+02	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	...	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	
mean	3.037183e+07	0.372583	14.127292	19.289649	91.969033	654.889104	0.096360	0.104341	0.088799	0.048919	...	16.269190	25.677223	107.261213	880.583128	0.132369	0.254265	0.272188	0.114606	
std	1.250206e+08	0.483918	3.524049	4.301036	24.298981	351.914129	0.014064	0.052813	0.079720	0.038803	...	4.833242	6.146258	33.602542	569.356993	0.022832	0.157336	0.208624	0.065732	
min	8.670000e+03	0.000000	6.981000	9.710000	43.790000	143.500000	0.052630	0.019380	0.000000	0.000000	...	7.930000	12.020000	50.410000	185.200000	0.071170	0.027290	0.000000	0.000000	
25%	8.692180e+05	0.000000	11.700000	16.170000	75.170000	420.300000	0.086370	0.064920	0.029560	0.020310	...	13.010000	21.080000	84.110000	515.300000	0.116600	0.147200	0.114500	0.064930	
50%	9.060240e+05	0.000000	13.370000	18.840000	86.240000	551.100000	0.095870	0.092630	0.061540	0.033500	...	14.970000	25.410000	97.660000	686.500000	0.131300	0.211900	0.226700	0.099930	
75%	8.813129e+06	1.000000	15.780000	21.800000	104.100000	782.700000	0.105300	0.130400	0.130700	0.074000	...	18.790000	29.720000	125.400000	1084.000000	0.146000	0.339100	0.382900	0.161400	
max	9.113205e+08	1.000000	28.110000	39.280000	188.500000	2501.000000	0.163400	0.345400	0.426800	0.201200	...	36.040000	49.540000	251.200000	4254.000000	0.222600	1.058000	1.252000	0.291000	

8 rows × 32 columns

In [17]:
```
train, test = train_test_split(df, test_size = 0.3)
trainX = train[df.columns[2:-1]]
trainY = train[df.columns[1]]

testX = test[df.columns[2:-1]]
testY = test[df.columns[1]]
```

In [18]:
```
gnb = GaussianNB()
```

In [20]:
```
clf = gnb.fit(trainX.values, trainY.values)
y_pred=clf.predict(testX.values)
print("Number of mislabeled points out pf a total %d points : %d"
      % (testX.shape[0], (testY.values !=y_pred).sum()))
```

```
Number of mislabeled points out pf a total 171 points : 9
```

In [21]:
```
testY
```

Out[21]:
```
380    0
300    1
389    1
140    0
526    0
      ..
40     1
295    0
362    0
206    0
304    0
Name: diagnosis, Length: 171, dtype: int64
```

In [22]:
```
y_pred
```

Out[22]:
```
array([0, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1,
       1, 0, 1, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 1,
       1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 0,
       1, 1, 1, 0, 0, 0, 1, 0, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0,
       0, 1, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 0,
       0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0])
```

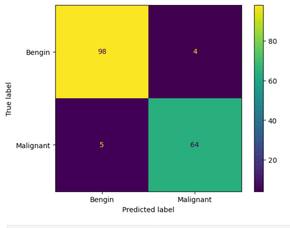
In [24]:
```
print('Accuracy of the NB Classifier is:',metrics.accuracy_score(y_pred,testY)*100)
```

```
Accuracy of the NB Classifier is: 94.73684210526315
```

In [27]:
```
cm=confusion_matrix(testY.values, y_pred)
cm_display = metrics.ConfusionMatrixDisplay(confusion_matrix = cm, display_labels = ['Bengin', 'Malignant'])

cm_display.plot()
plt.show
```

Out[27]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



In [32]:
```
sample = [[15.78,17.89,103.6,781,0.0971,0.1292,0.09954,0.06606,0.1842,0.06082,0.5058,
0.9849,3.564,54.16,0.005771,0.04561,0.02791,0.01282,0.02058,0.004144,20.42,136.5,1299,0.1396,0.5609,0.3965,0.181,0.3792,0.1048]]
pred = clf.predict(sample)
res=pred[0]
if res==0:
    print('Bengin')
if res==1:
    print('Malignant')
```

```
Malignant
```

In [34]:
```
sample = [[11.94,18.24,75.71,437.6,0.08261,0.04751,0.01972,0.01349,0.1868,0.0611,
           0.6329,1.52,17.47,0.00721,0.00838,0.01311,0.008,0.01996,0.002635,13.1,21.33,83.67,527.2,0.1144,0.08906,0.09203,0.06296,0.2785,0.07408]]
pred = clf.predict(sample)
res=pred[0]
if res==0:
    print('Bengin')
```

```
if res==1:
    print('Malignant')
```

```
Bengin
```

In [ ]: